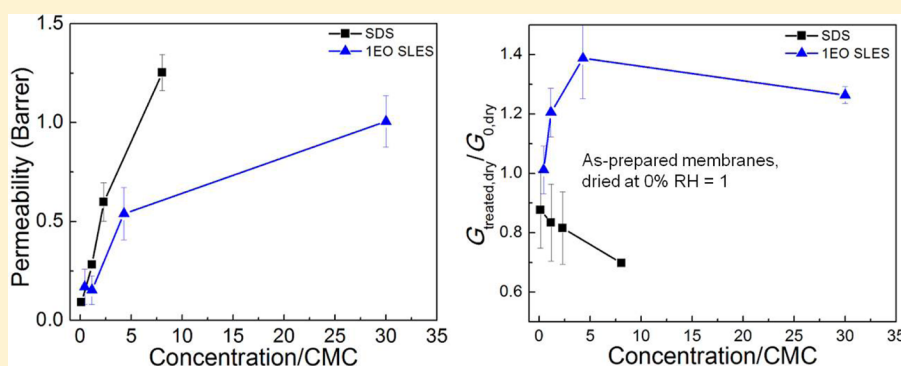


Effects of Anionic Surfactants on the Water Permeability of a Model Stratum Corneum Lipid Membrane

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Supporting Information



ABSTRACT: The stratum corneum (SC) is the outermost layer of the epidermis and has a brick-and-mortar-like structure, in which multilamellar lipid bilayers surround flattened dead cells known as corneocytes. The SC lipid membranes provide the main pathway for the transport of water and other substances through the SC. While the physicochemical properties of the SC can be affected by exogenous materials such as surfactants, little is known about how the water barrier function of the SC lipid membranes is compromised by common surfactants. Here, we study the effect of common anionic surfactants on the water permeability of a model SC lipid membrane using a quartz crystal microbalance with dissipation monitoring (QCM-D). Particularly, the effect of sodium dodecyl sulfate (SDS) and sodium lauryl ether sulfate (SLES) is compared. These two surfactants share commonality in their molecular structure: sulfate in the polar headgroup and the same apolar tail. The mass of the lipid membranes increases after the surfactant treatment at or above the critical micelle concentration (CMC) of the surfactants due to their absorption into the membranes. The incorporation of the surfactants into the lipid membranes is also accompanied by partial dissolution of the lipids from the model SC lipid membranes as confirmed by Fourier-transform infrared (FT-IR) spectroscopy. Although the water sorption of pure SDS is much lower than that of pure SLES, the water sorption of SDS-treated membranes increases significantly similar to that of SLES-treated membranes. By combining QCM-D and FT-IR spectroscopy, we find that the chain conformational order and stiffness of the lipid membranes decrease after SDS treatment, resulting in the increased water sorption and diffusivity. In contrast, the conformational order and stiffness of the SLES-treated lipid membranes increase, suggesting that the increased water sorption capacity of SLES-treated lipid membranes is due to the hygroscopic nature of SLES.

INTRODUCTION

The outermost layer of the skin, the stratum corneum (SC), is composed of flattened dead cells called corneocytes that are embedded in an intercellular matrix of stacked lipid lamellae consisting predominantly of free fatty acids, ceramides, and cholesterol.^{1–3} The SC lipid lamellae are thought to be the main pathway for water transport through the SC and play a significant role in regulating the loss and uptake of water through the skin.⁴ The physicochemical properties of the SC are affected by exogenous substances applied to the skin such as cosmetic, dermatological, and/or cleansing products. Most cleansing products contain surfactants as foaming agents that could potentially compromise the barrier properties of the SC. For example, it was shown that surfactants in solution induce

swelling of an isolated SC, a reduction in the elastic modulus of the SC, and the decomposition of the SC into individual corneocytes.^{5–7} Especially, high levels of swelling were induced by anionic surfactants, whereas the swelling of the SC by cationic and nonionic surfactants was negligible.⁶ Anionic surfactants such as sulfate-based surfactants, nevertheless, are widely used in personal care products such as soaps, shampoos, and toothpastes because they are effective cleaners.⁸ Numerous studies in this field have studied the effects of anionic surfactants on the physicochemical properties of the SC and/

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or the SC lipid membranes.^{7,9–11} Although useful information on surfactant-induced changes in the mechanical properties of the SC,¹² the lamellar structure and ordering of lipids,^{13–15} and the dissolution properties of lipid membranes^{16–18} has been accumulated, little is known about how water permeation through the SC lipid membranes is compromised by anionic surfactants.

In this work, we study the effect of commonly used anionic surfactants on the water permeability of a model stratum corneum (SC) lipid membrane composed of an equimolar mixture of ceramide, palmitic acid, and cholesterol using a quartz crystal microbalance with dissipation monitoring (QCM-D). In particular, we study how sodium dodecyl sulfate (SDS) and sodium lauryl ether sulfate (SLES) with one ethoxy (1EO) group modify the water permeation properties of a model SC lipid membrane. By taking advantage of the dissipation monitoring capability of QCM-D, humidity-induced changes in the mass and mechanical properties of SC lipid membranes before and after treatment with each surfactant are obtained. It is shown that although the moisture sorption capabilities of the two surfactants are very different, the water solubility of surfactant-treated lipid membranes shows a similar trend regardless of the surfactant. On the other hand, the water diffusivity of SDS-treated lipid membranes is more significantly increased than that of SLES-treated lipid membranes. The structural changes in the surfactant-treated lipid membranes as analyzed using Fourier transform infrared (FT-IR) spectroscopy provide insights into the physical reason behind the observed difference in the water diffusivity, and, in turn, the water permeability of these surfactant-treated SC lipid membranes. We believe that combining the two complementary characterization techniques used in this work (QCM-D and FT-IR) can be extremely versatile in establishing the structure–property relationship of other types of soft materials after surfactant treatments.¹⁹

MATERIALS AND METHODS

Materials. Model SC lipid membranes are formed using a 1:1:1 (mol/mol/mol) mixture of ceramide (CER), free fatty acids (FFA), and cholesterol (CHOL). CER (Brain, Porcine) is purchased from Avanti Polar Lipids. According to the supplier, the brain ceramide is predominantly *N*-(octadecanoyl)-sphing-4-enine (>99%). Palmitic acid (C16), used as the saturated FFA component, and CHOL are purchased from Sigma-Aldrich. Each component is initially dissolved in the mixture of hexane and ethanol (2:1 v/v) and an equimolar mixture of three lipid components is prepared to a concentration of 4.5 mg/mL using the same solvent mixture. The equimolar ratio of these three components is known to be similar to the physiologic ratio in the SC lipid membrane²⁰ and has been used as model SC lipid membranes in many other studies.^{21–23} Sodium dodecyl sulfate (SDS) is purchased from Sigma-Aldrich. Sodium lauryl ether sulfate (SLES) with one ethoxy (1EO) group is purchased from Stepan Company. The molecular structure of each surfactant is shown in Figure S1. The critical micelle concentrations (CMC) of SDS and SLES are reported to be 8.7 mM¹⁰ and 0.7 mM,²⁴ respectively. The surfactants are used as received. SDS and SLES solutions are dissolved in deionized (DI) water and used without pH titration. The pH of both SDS and SLES solutions are about 6.5.

Preparation of Model SC Lipid Membranes. Prior to membrane deposition, gold-coated quartz crystals are cleaned in H₂O/H₂O₂/NH₄OH mixture (5:1:1 v/v/v) at 75 °C for 30 min, then rinsed with deionized (DI) water thoroughly, and dried with compressed air. Lipid membranes are deposited onto the cleaned gold-coated quartz crystals using an airbrush (PEAK C-5, Bear Air, Japan) connected to compressed nitrogen.²⁵ The mass of the lipid membranes on the crystal is controlled by varying the amount of

sprayed lipid solution. Deposited lipid membranes are vacuum-dried for over 2 h at room temperature and are annealed at 75 °C under water-saturated atmosphere for 10 min. This technique is known to generate homogeneous, tightly packed multilamellar lipid membranes that are structurally reminiscent of in vivo SC lipid membranes.^{21,25,26} Moreover, the thickness of stacked membranes can be easily varied by using different amounts of lipid mixture solution for spray coating.

Surfactant Treatment. The SC lipid membranes are immersed in each surfactant solution (about 15 mL) for 10 min followed by gently rinsing in DI water (about 30 mL) for 20 s to remove excess surfactant molecules on the surface of the lipid membranes. Then, the surfactant-treated lipid membranes are dried in ambient condition until excess water on the surface disappears. Before reloading the QCM crystals, the backside of crystals are gently but thoroughly cleaned with cotton swab using DI water. We have tested and found that the remounting of QCM crystals back to the QCM module has a negligible effect on the accuracy of thickness and shear modulus measurements with $\pm 0.1\%$ and $\pm 0.7\%$ standard deviation, respectively.

Quartz Crystal Microbalance with Dissipation Monitoring (QCM-D). An E4 QCM-D unit (Q-Sense, Inc.) is used to obtain the mechanical properties and mass changes of the model SC lipid membranes by measuring shifts in the frequency and dissipation of quartz crystals coated with lipids. The baseline frequency and dissipation are first recorded with a dry-state blank crystal by flowing dry N₂ at a flow rate of 100 sccm. After the deposition of lipid membranes on the QCM sensors, the sensors are loaded in the QCM chambers. For humidity-dependent studies, the frequency and dissipation are monitored at varying RH by flowing a mixture of dry and humid N₂. The relative humidity (RH) in N₂ stream is precisely controlled by adjusting the ratio of dry (0% RH) and water-saturated (100% RH) N₂. The total flow rate of the mixture is fixed at 100 sccm, and the temperature in the QCM chambers is maintained at 25 °C. The lipid membranes in the QCM chambers are initially dried under 0% RH N₂ flow for about 5 h and then exposed sequentially to 20, 40, 60, 80, and 100% RH for 2 h each (see Figure S2a). The measured frequency and dissipation shifts from multiple overtones are then fitted using the Voigt viscoelastic model incorporated in Q-Sense analysis software (QTools) to obtain the mass and shear modulus of the lipid membranes (see Figure S2b). During the modeling, the density of the lipid membranes is assumed to be 1000 kg/m³. All QCM-D results throughout this work are obtained by averaging three independent measurements.^{27–29}

Fourier Transform Infrared (FT-IR) Spectroscopy. A Nicolet 8700 FT-IR spectrophotometer (Thermo Scientific) is used to obtain the FT-IR spectra of surfactant-treated lipid membranes on ZnSe windows (Phoenix Infrared, MA). Lipid membranes are prepared on ZeSe windows using the same protocol as described above. The size of the ZeSe window is 25.4 mm in diameter and 1.0 mm in thickness. To ensure that we monitor the effect of surfactant concentration on a single SC lipid membrane, small aliquots of surfactant solutions of different concentrations are placed on top of different regions of the single lipid membrane. The surfactant-treated lipids are rinsed in DI water for 20 s. The spectra are taken by averaging 64 scans at 2 cm^{−1} or 0.06 cm^{−1} wavenumber resolution.

Determination of Diffusivity, Solubility, and Permeability. The water diffusivity (*D*) of the SC model lipid membranes is obtained from the plot of mass change as a function of time under 95% RH using a one-dimensional diffusion equation as follows:

$$\frac{m_t - m_0}{m_\infty - m_0} = 1 - \sum_{n=0}^{\infty} \frac{8}{(2n+1)^2 \pi^2} \exp \left[\frac{-D(2n+1)^2 \pi^2 t}{l^2} \right] \quad (1)$$

where, *m_t*, *m₀*, and *m_∞* are the mass of the SC lipid membranes at time = *t*, 0, and ∞, respectively. *l* and *t* are the thickness of the dried lipid membranes and time, respectively. *D* is diffusion constant. The above equation can be simplified by taking the first term of the series on the right-hand side of the equation:

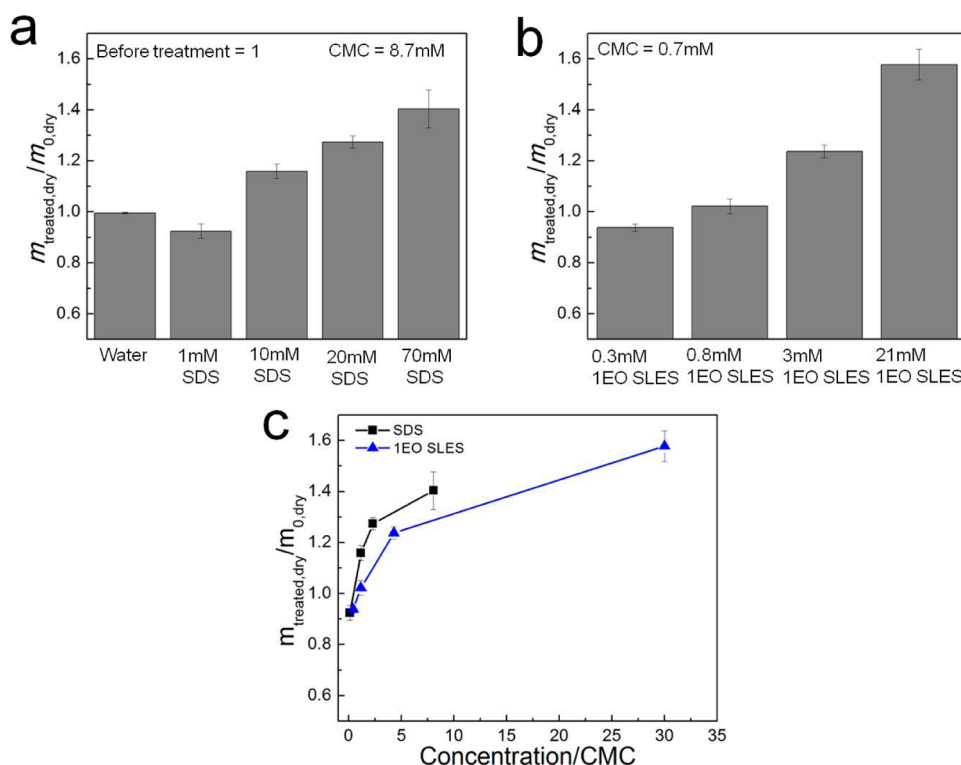


Figure 1. Relative mass of dried lipid membranes after treatment with (a) DI water and SDS and (b) 1EO SLES. (c) Relative mass of dried lipid membranes as a function of surfactant concentration normalized by the critical micelle concentration (CMC) of each surfactant. $m_{0,\text{dry}}$ and $m_{\text{treated,dry}}$ denote the dried masses of as-prepared lipid membranes and surfactant-treated lipid membranes, respectively.

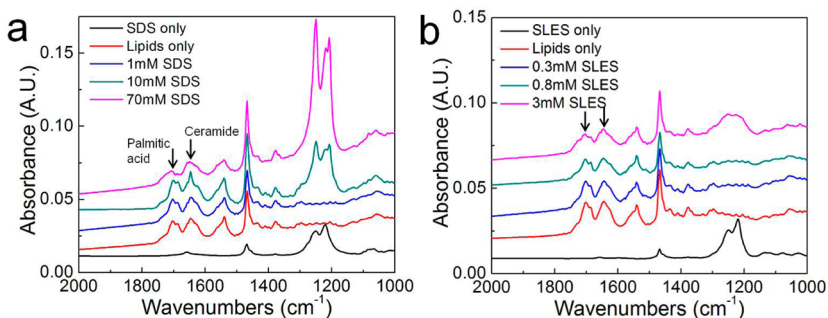


Figure 2. FT-IR spectroscopy of model SC lipid membranes after treatment with (a) SDS and (b) SLES. Data measured from nontreated lipid membranes and each surfactant is included in the corresponding panels.

$$-\frac{1}{\pi^2} \ln \left[\frac{(m_\infty - m_t)\pi^2}{8(m_\infty - m_0)} \right] \times l^2 = Dt \quad (2)$$

D is obtained by fitting eq 2 to the water sorption curves obtained by QCM-D within the 35–85% sorption range; the equation has been shown to apply with a high accuracy for this range (see Figure S3). The solubility (S) of water is determined by the change in the mass of the lipid membranes between dry (0% RH) and wet (95% RH) conditions, $m_\infty - m_0$. The permeability (P) of water in SC lipid membranes is deduced using the obtained values of D and S , as described by $P = D \cdot S$.

RESULTS AND DISCUSSION

The mass change of the model SC lipid membranes after their exposure to surfactant solutions shows significant absorption of the surfactants into the lipid membranes. As a control experiment, we confirm that there is essentially no change in the mass of dried lipid membranes after the membranes are immersed in DI water for 30 min (Figure 1a). Interestingly, the

relative mass changes of dried SC lipid membranes after immersion in surfactant solutions share a common behavior when the surfactant concentrations are normalized by the critical micelle concentration (CMC) of each surfactant as shown in Figure 1a and b. In general, the mass of dried lipid membranes after treatment with surfactant slightly decreases at concentrations below CMC; 7.5 ± 2.9 and $6.2 \pm 1.4\%$ decreases are observed for model SC lipid membranes immersed in 0.12 and $0.43 \times \text{CMC}$ of SDS and SLES solution, respectively. However, when model SC lipid membranes are immersed in the surfactant solutions at around their CMCs, the mass of the lipid membranes increases, indicating that surfactant molecules are incorporated into these lipid membranes; 15.9 ± 2.9 and $2.2 \pm 2.9\%$ increases are observed in the SDS- and SLES-treated lipid membranes at 1.15 and $1.14 \times \text{CMC}$ of each surfactant, respectively. The mass of dried lipid membranes further increases after treatment at higher surfactant concentrations. Previously, a similar behavior was

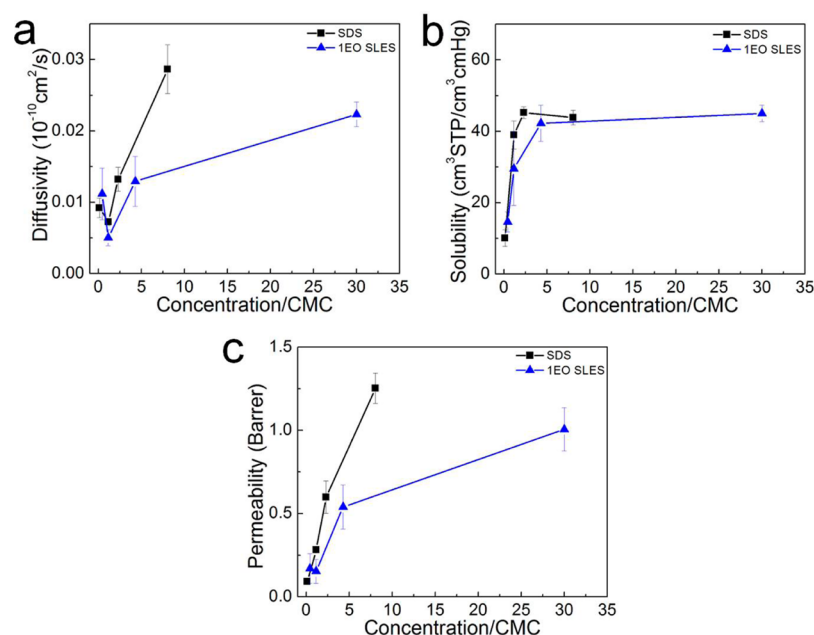


Figure 3. (a) Water diffusivity, (b) solubility, and (c) permeability of SC model lipid membranes after the treatment with SDS and SLES as a function of surfactant concentration normalized by CMC. 1 Barrer is $10^{-10} [\text{cm}^3 (\text{STP}) \cdot \text{cm}] / [\text{cm}^2 \cdot \text{s} \cdot \text{cmHg}]$. In SI units, 1 Barrer is approximately $7.5 \times 10^{-18} [\text{m}^3 (\text{STP}) \cdot \text{m}] / [\text{m}^2 \cdot \text{s} \cdot \text{Pa}]$.

observed in a QCM-based SDS-injection study in which the mass of a model SC lipid membrane decreased slightly when the concentration of the surfactant was below its CMC and then increased when the concentration was raised above the CMC.²³ Moreover, in a surfactant study using epidermis, it was shown that only micellar SDS contributes to the penetration of SDS molecules into the epidermis.³⁰ In line with these previous studies, our observation indicates that the incorporation of surfactant molecules into lipid membranes occurs at and above CMC, suggesting a critical role of micelles in interacting with the model SC lipid membrane. The incorporation of both SLES and SDS was shown to be fast (~ 10 s),³¹ suggesting that the difference in the incorporation of the two surfactants into phospholipid membranes was likely due to a thermodynamic origin rather than a kinetic one. It was suggested that the increased hydrophilicity and/or molecular size due to the presence of EO groups hinder the incorporation of SLES into these phospholipid membranes.

FT-IR spectroscopy is used to probe the compositional changes that take place upon the exposure of the model SC lipid membranes to surfactant solutions. In particular, we are interested in investigating whether lipids are extracted while surfactants are being added to the lipid membranes. The FT-IR spectra of nontreated lipid membranes and lipid membranes immersed in surfactant solutions below CMC show no discernible difference as shown in Figure 2. However, in the spectrum of lipid membranes treated with SDS at its CMC, the appearance of asymmetric S–O stretching band (a broad doublet at $1200\text{--}1300 \text{ cm}^{-1}$) from SDS (clearly seen in Figure 2a)^{32,33} indicates a substantial amount of SDS molecules incorporated in the membranes, which is consistent with the increase in mass detected by QCM-D ($15.9 \pm 2.9\%$ in Figure 1c). However, the same peaks are not clearly observed in the lipid membranes treated with SLES at its CMC, indicating that the incorporation of surfactants is not significant. This observation again is consistent with the very small increase in the mass of SLES-treated lipid membranes at the corresponding

concentrations ($2.2 \pm 2.9\%$ increase for SLES treatment in Figure 1c). However, at concentrations above CMC, the spectra of all surfactant-treated lipid membranes show asymmetric S–O stretching peaks from each surfactant, indicating that the mass increase of the lipid membranes treated at these concentrations is indeed induced by the incorporation of surfactants into the model SC lipid membranes.

Interestingly, a slight decrease in the intensity of the peaks for palmitic acid ($\sim 1700 \text{ cm}^{-1}$, C=O stretching band of the carboxyl group),^{34,35} and ceramide ($\sim 1650 \text{ cm}^{-1}$, Amide I)³⁶ is observed in the lipid membranes immersed in surfactant solutions above CMC. A peak for cholesterol that does not overlap with the peaks of palmitic acid and ceramide is located at about 1060 cm^{-1} .³⁴ However, it is difficult to identify the relative change of the peak intensity from cholesterol because of overlaps with peaks from surfactants. Nevertheless, our observation indicates that lipid molecules in the membranes are partially extracted while surfactants molecules are being absorbed into the membranes. Moreover, a preferential depletion of palmitic acid by the two surfactants is observed when the ratio (R) of the peak area (A) for palmitic acid (Pal) to that for ceramide (Cer) ($R = A(\text{Pal})/A(\text{Cer})$) is compared (Figure S4). The ratios of peak areas (R) before and after surfactant treatment ($\sim 4.5 - 8 \times \text{CMC}$) are 0.52 and 0.85 for SDS- and SLES-treated lipid membranes, respectively, indicating preferential removal of palmitic acid over ceramide in both cases.

The partial dissolution of lipids and the incorporation of surfactants in the lipid membranes are likely to significantly affect their water permeability and barrier properties. We study the effect of these surfactants on the water diffusivity (D), solubility (S), and permeability (P) of the model SC lipid membranes. It is important to remember that for water permeation to occur, two phenomena must occur: water must get absorbed into the membrane from the air phase, and subsequently it has to diffuse across the membrane. Diffusivity

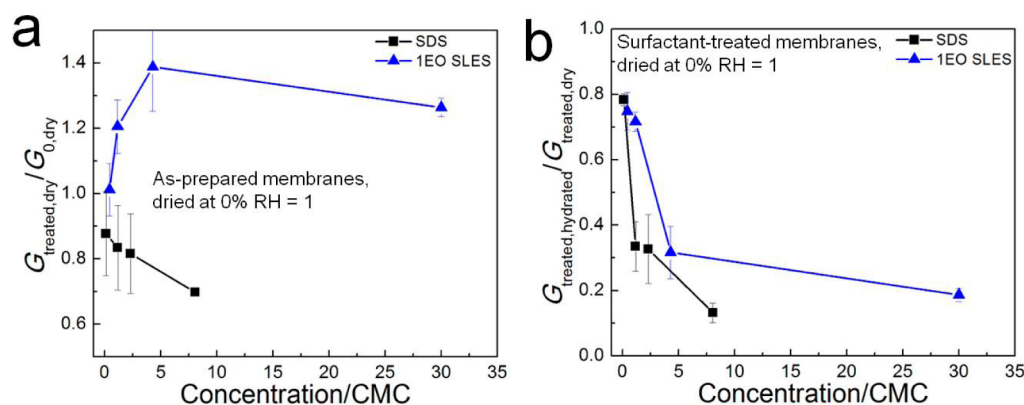


Figure 4. Normalized shear modulus of SC model lipid membranes treated with surfactants at (a) 0 and (b) 100% RH. The reference state in (a) and (b) is the shear modulus of dried membranes after no treatment (as-prepared) and treatment with each surfactant, respectively. $G_{0,\text{dry}}$, $G_{\text{treated,dry}}$, and $G_{\text{treated,hydrated}}$ denote the shear moduli of as-prepared (nontreated) lipid membranes at 0% RH, surfactant-treated lipid membranes at 0% RH, and surfactant-treated lipid membranes at 100% RH, respectively.

is a kinetic parameter that illustrates how fast water diffusion occurs through the lipid membranes at a given RH, whereas solubility is a thermodynamic parameter, which shows how much water can be absorbed in a given lipid membrane at a given RH. Therefore, the water permeability of surfactant-treated lipid membranes is deduced using $P = D \cdot S$, which is affected by both kinetic (D) and thermodynamic (S) factors.

In general, the diffusivity of the membranes increases after immersion in the surfactant solutions. The increase in the diffusivity of the SDS-treated lipid membranes, however, is more significant than that of the SLES-treated lipid membranes at concentrations above CMC as seen in Figure 3a. The solubility of these lipid membranes after immersion in the surfactant solutions shows a similar increase as shown in Figure 3b. A sharp increase in solubility by a factor of 4–5 at high surfactant concentrations suggests that the surfactant incorporation significantly contributes to the increase in the water solubility of lipid membranes. The SDS-treated lipid membranes have higher permeability than SLES-treated membranes, which is caused by the large diffusivity increase. However, considering the similar trends observed in mass increase and compositional changes in these surfactant-treated lipid membranes, it is not clear what causes the significant increase in the diffusivity of SDS-treated lipid membranes. In addition, since the water sorption capacities of a pure SDS film and as-prepared lipid membranes are much lower than those of a pure SLES film (see Figure S5), the similar level of increase in the water solubility in Figure 3b seems somewhat counterintuitive. The rest of this study focuses on finding which factors lead to the substantial increase in the water diffusivity, and thus the permeability, of SDS-treated lipid membranes.

Since the structural integrity of lipid membranes is known to be critical in the water permeation of the SC,³⁷ we investigate how the mechanical properties of lipid membranes, one indicator of membrane integrity,^{38,39} are affected after surfactant treatments by taking advantage of the dissipation monitoring capability of QCM-D. Figure 4 shows the relative shear modulus of surfactant-treated lipid membranes plotted as a function of surfactant concentration normalized by CMC. To exclude the effect of water in lipid membranes, the shear moduli of lipid membranes dried at 0% RH after no treatment (reference state) and surfactant treatments are first compared (Figure 4a). Interestingly, the shear modulus of the SDS-

treated lipid membranes decreases monotonically as a function of surfactant concentration, whereas that of the SLES-treated lipid membranes initially increases with increasing surfactant concentration and appears to reach a plateau at high surfactant concentrations.

Next, we compare the relative shear modulus of the fully hydrated surfactant-treated lipid membranes to that of the fully dried membranes after treatment with each surfactant (reference state) as a function of normalized surfactant concentration (Figure 4b). The shear moduli of all lipid membranes decrease significantly in the presence of water, indicating that water molecules play an important role in softening these surfactant-treated lipid membranes. The SDS-treated lipid membranes show a sharp decrease in shear modulus. This significant reduction in the shear modulus of the SDS-treated lipid membranes is likely related to the reduction observed in the dry membranes (Figure 3a). This result is surprising because, as briefly mentioned earlier, the water sorption capability of pure SDS is quite low. The possible mechanism of this somewhat counterintuitive observation is provided below.

To rationalize the changes in the mechanical properties as well as the diffusivity of the surfactant-treated lipid membranes, we closely examine the FT-IR spectra of the surfactant-treated membranes. In particular, the shifts of asymmetric C–H₂ stretching ($\sim 2917\text{ cm}^{-1}$) and symmetric C–H₂ stretching ($\sim 2849\text{ cm}^{-1}$) peaks are investigated. It has been reported that the frequencies of the asymmetric and symmetric C–H₂ stretching peaks in the SC lipid membranes and phospholipids increase upon melting and high RH.^{40–42} These observations suggest that C–H₂ stretching peaks are indicators for the chain conformational order of the lipids. In the present study, the increase in the frequencies of both peaks is observed in the SDS-treated lipid membranes, whereas the frequencies of the two peaks decrease in the SLES-treated lipid membranes (Figure 5; see FT-IR spectra in Figure S6). These results indicate that the chain conformational order of the SDS-treated lipid membranes is decreased upon surfactant absorption, whereas that of SLES-treated lipid membranes is increased. These results are consistent with the changes of the shear moduli of these surfactant-treated lipid membranes since less chain conformational order in lipids and surfactants is likely to result in softer membranes. We note that the FT-IR spectra of pure SDS and SLES show that pure SDS is more ordered than SLES (the symmetric and asymmetric CH₂ stretching peaks of

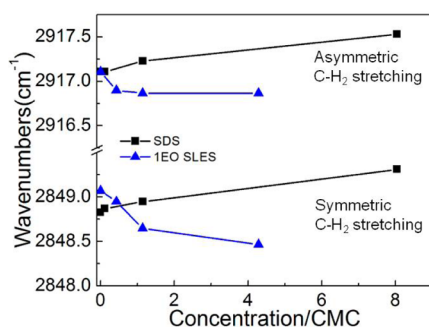


Figure 5. FT-IR wavenumber shifts of asymmetric C–H₂ stretching and symmetric C–H₂ stretching peaks.

pure SDS are located at 2849.67 and 2917.15 cm^{−1}, respectively, whereas those peaks for pure SLES are 2850.81 and 2918.69 cm^{−1}, respectively). Such a result excludes the possibility that the lower frequency observed for the SLES-treated SC lipid membranes are due to the more ordered SLES compared to SDS.

The FT-IR analysis of the surfactant-treated lipid membranes sheds some light on the changes we have observed in the permeability as well as the mechanical properties of these model SC lipid membranes. Previously, an IR spectroscopy-based study has shown that the water permeability of excised SC is strongly correlated with the chain conformational order of lipids.⁴³ Also, previous studies based on X-ray scattering have shown that the lamellar phases of lipids in the SC are destroyed by SDS treatment.⁴⁴ Our study using FT-IR shows that the chain order of the model SC lipid membranes is disrupted by SDS, which likely is responsible for the significant decrease in the shear modulus as well as the drastic increase in the water diffusivity in the SDS-treated lipid membranes. The large enhancement of the water solubility in the SDS-treated lipid membranes could also be attributed to the significant disorder present in the SDS molecules within the membranes. In contrast, the increase in the water solubility of the SLES-treated lipid membranes is likely due to the high water sorption capacity of SLES itself (Figure S5).

CONCLUSIONS

We have studied the effect of SDS and SLES on the water permeability of a model stratum corneum (SC) lipid membrane using a quartz crystal microbalance with dissipation monitoring (QCM-D) and Fourier-transform infrared (FT-IR) spectroscopy. The two surfactants incorporate into the model SC lipid membrane when the concentration of the surfactant solution is higher than critical micelle concentration (CMC). The incorporation of these surfactants into the lipid membranes is accompanied by a partial loss of the lipids. The water diffusivity of SDS-treated lipid membranes shows more significant increase compared to that of SLES-treated lipid membranes. Also, the modulus of SDS-treated lipid membranes is lower than that of SLES-treated ones regardless of hydration states of the membranes, indicating that SDS has a different effect on the structure and properties of the model SC lipid membranes than SLES. These observations, we believe, are due to the loss of conformational order in the lipid membranes when SDS is incorporated, as determined by FT-IR analysis. Our results provide fundamental insight into how the water permeation of SC lipid membranes can be modified by these anionic surfactants. Also, our study emphasizes the importance of

using an integrated approach to study the water permeability of SC lipid membranes by establishing the structure–property relationship using complementary characterization methods (i.e., QCM-D and FT-IR spectroscopy). The material exchange and the corresponding property change of SC lipid membranes observed in this study will be helpful in understanding the surfactant–lipid interactions as well as the interactions between surfactant and soft matter in general.⁴⁵ We anticipate that our approach will also be useful for the development of surfactant-based drug delivery and antibacterial systems in which surfactant–lipid (plasma membrane) interactions are critical.⁴⁶ Future studies on the effect of other (nonionic, cationic and/or amino acid-based) surfactants on the model SC lipid membranes using our approach will be beneficial to the development of surfactant-based skin products.

ASSOCIATED CONTENT

Supporting Information

Molecular structure of surfactants, frequency and dissipation shifts in lipid membranes and corresponding thickness and shear modulus, diffusivity (*D*) obtained by eq 2, humidity-dependent relative mass change of pure surfactants, calculation of preferential depletion of palmitic acids, and FT-IR spectra in the range of asymmetric C–H₂ stretching and symmetric C–H₂ stretching frequencies. These materials are available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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